IE University

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Classifying Book Content For Age Suitability

A Multi-feature Machine Learning Approach to Automated Age-Appropriateness Classification

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Abstract

This project develops a machine learning classification system capable of automatically assigning age suitability labels to books based on its content. A balanced dataset of 8,000 book descriptions drawn from Goodreads, as well as supplementary open-source repositories, the project extracts 23 linguistically motivated features covering 5 analytical domains: readability, linguistic structure, sensitivity markers, sentiment, and stylistic register. With the help of 5 classical machine learning classifiers, after training and evaluation, with a support vector machine (SVM) the model achieved a test accuracy of 75.1%. Although this falls below the 80% original target, the results still improve a single feature baseline (43.7%). This engineered feature set provides transparent, human-interpretable, justification for each decision. A fully functional desktop application is developed, allowing users to paste any text and receive an instant verdict. Which of course is accompanied with a confidence score and an explanation of the top feature contributions driving the result. This project addresses a genuine societal need, supporting parents, educators and publishers in navigating the increasingly complex world of digital and physical reading materials.

Keywords: Age Suitability, Text Classification, Readability, Natural Language Processing, Machine Learning, Content Moderation, Children's Literature, Explainable AI

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1. Introduction

1.1 Background and Motivation

There is a recurring question between parents, educators, librarians, publishers… What is appropriate reading material for a given age group? Children and teenagers have greater access than ever to books, e-reads and online platforms, this requires a level of control over what is appropriate and what is not which is unprecedented. A suitable book for a 12 year old may contain themes, language or scenarios which are inappropriate for a 7 year old but may feel patronising to a 16 year old.

For years, age labelling has relied solely on expert judgment, librarians, teachers and publishers apply their knowledge to assign a recommended age or reading level. The benefit from this is the human insight and cultural sensitivity, it suffers from inconsistency and it can be considered slow for large catalogues. As digital reading platforms continue to grow and self publishing markets do so as well. The gap between age labelling and supply of human classifications has grown apart.

Automated text analysis offers a promising solution to this problem. Advances in natural language processing (NLP) and machine learning have made it possible to extract meaningful signals from text at scale. Flesch-Kincaid Grade Level and other formulaic measures of readability of written texts have been in common use for decades but do not assess the thematic appropriateness of a given text. An effective age-suitability classifier should assess not just sentence length and other superficial aspects of vocabulary complexity, but also mature themes, emotional substance, and details of violence, swearing, drugs, and so on.

We propose and present an internal evaluation of a multi-feature based machine-learning model that classifies books into four age groupings: +4 (early readers), +10 (middle grade), +12 (young adult), and +18 (adult). The four age groupings chosen for classification in this work are a reflection of a real-world need observed in the literature. While research has focused on the binary problem of separating children's books from adult books, the intermediate age groupings (+10 and +12) remain under-explored. The distinction between childhood and adolescence is particularly important because of the wide-ranging development of the brain in pre-adolescence and early adolescence.

1.2 Problem Statement

Given a passage of raw text from a book, can we determine with good accuracy with a machine learning model which age category the book belongs to (4, 10, 12, 18), while keeping a human-understandable, interpretable and practically usable model?

This problem has several sub-components. The first is the feature engineering problem. Which text features best predict the age categories? The +4 and +18 categories can be distinguished by simpler vocabulary, syntax, and more positive emotional tone and story concepts in the +4 books. The +18 titles have more complex sentences and vocabulary, and they are the most likely

to touch on delicate topics. The differences between +10 and +12 readers were modest. The linguistic complexity of the books for both +10 and +12 readers was similar, but they vary in the relevance of identity topics, romantic topics, and mild conflict typical of young adult novels.

Second, there is the modeling problem: which classification model is best suited for this multi-class classification, tabular-feature data? Classical machine learning models such as logistic regression, support vector machines, random forests, and gradient increasing all have different tradeoffs in terms of accuracy, interpretability, and robustness. Additionally, another important factor of this project, besides the model's performance as an accurate classifier, is the ability of the model to explain its classification decision, a feature that can increase trust of the model by parents and teachers, respectively.

Third, there is a deployment problem. In what form can we ensure that this classifier is used in a user-friendly way by a non-programmer? In practical terms, a desktop application that allows one to paste some text and generates an immediate human-readable verdict would give this research apparent utility.

1.3 Research Question

The primary research question guiding this project is:

> ***How accurately can book text be used to predict age-appropriateness, and which linguistic, semantic, and content-sensitivity features most strongly determine suitability across age categories such as +4, +10, +12, and +18?***

First, to address this question we ask two subsidiary questions: (1) Do the readability-based measures provide sufficient discrimination between the age categories by themselves? (2) How much predictive power is contributed by the sensitivity-based measures,over and above that provided by the readability-based measures? (3) Which of the five feature groups defined above provides the strongest age-discriminative signal? (4) What is the performance ceiling of classical ML approaches on a feature set of this kind?

1.4 Objectives

The project has four main objectives:

1. A multi-sourced balanced dataset of 8000 book text excerpts was developed and evaluated, each labeled with one of 4 ages (4, 10, 12, 18). The Goodreads corpus was mostly used, supplemented with open-source children's and middle grade story collections.
2. 23 features across five domains of analysis: readability, language structure, sensitivity markers, sentiment, and stylistic register.
3. Train and test five classical machine learning classifiers, selecting the best performing, and provide a full analysis of feature importance and per-class performance.
4. A functional desktop application was built to allow any user to paste in book text and receive an instant age-classification result with probabilities & feature-level explanations.

A primary measurable objective of this project was to achieve classification performance greater than 80% accuracy in the held-out test set, with a sub-objective of establishing a statistically important improvement over the naive single-feature baseline.

1.5 Report Structure

The report is organized as follows. A literature review is presented in section 2, focusing on research in the area of readability prediction, automatic content classification, and age suitability prediction. Section 3 describes the data collection and processing steps, while section 4 describes the feature engineering process. Section 5 performs exploratory data analysis to show that the engineered features are able to separate each of the four age categories. Section 6 discusses modeling experiments and model evaluation. Section 7 describes the prototype's applications, followed by its results, limitations, and ethical considerations in section 8. Section 9 concludes with a summary and suggestions for future research, then references and appendices follow.

With the problem in sight, the objectives established and the target accuracy set, the next section reviews the existing literature on readability, content moderation, and age classification that informed the design choices made throughout this project.

2. Literature Review

Automating the classification of books according to their expected age of interest arguably lies at the nexus of three active fields in the literature: computational approaches to estimating reading levels, computational approaches to content moderation using natural language processing (NLP), and educational data mining. We discuss relevant existing research from simple readability measures to multi-dimensional classification systems.

2.1 Readability and Linguistic Complexity

The readability formula is the earliest and most widely-used mechanism to measure the readability of a text. The Flesch Reading Ease formula was published in 1948. The formula combines average sentence length and average number of syllables per word into a single number with a range from 0 (very difficult) to 100 (very easy). Closely related to this is the Flesch-Kincaid Grade Level formula, which expresses the same two underlying parameters in terms of a US school grade level, which teachers find easier to use. Both formulas have been extensively validated, and are still used by educational publishers today.

Subsequent readability formulas have used this approach in different ways. Gunning, R. (1952) included a similar experimental approach by using the percentage of polysyllables. The Dale-Chall Readability Formula uses a list of 3000 easy words to compare the text which is being analyzed against words that are known by most people. Conversely, the SMOG Index, by McLaughlin, G. H. (1969), which uses only the number of polysyllabic words, is also a very popular metric in health communications research. While these three formulas thus capture somewhat different aspects of textual complexity, as this project has demonstrated, the combined signal from these three measures is much more discriminative.

For example, in a comparison between different classification and regression approaches for predicting student performance based on linguistic features (such as those used for readability prediction), Strecht et al. (2015) found a strong correlation between vocabulary richness and density of content words and academic performance. They inferred that the linguistic features correlate with the text's cognitive load required from the reader. This finding provides direct support for including vocabulary richness (type-token ratio) and content word ratio in the set of features explored in the present project.

From a practical perspective, the trend in the Flesch Reading Ease score in the present data set is strong. +4 books have a mean Reading Ease score of 88.1 (very easy); +18 books have a mean Reading Ease of 51.6. The monotonic decrease by age group suggests that readability alone should potentially be a strong first-order predictor. However, the results of Section 5 suggest that the gap between +10 and +12 books is much smaller than the gap between +4 and +10, and we may need more dimensions.

2.2 Semantic Mapping and Thematic Maturity

Besides the complexity metrics based on raw linguistic features, Brewer and Ng (2019) proposed a novel framework for age-suitability prediction, using deep learning, in the paper 'Age-Suitability Prediction for Literature Using a Recurrent Neural Network Model', which was also presented at the IEEE International Conference on Tools with Artificial Intelligence (ICTAI). They showed that recurrent neural networks trained on raw text embeddings could achieve much better classification performance than feature-based models, given a suitable amount of labeled training data. The main contribution of the work was showing that the 'weight' of thematic content, as measured by distributed word representations, provides more discriminative information than readability formulas do.

A follow-up to this paper was published in Brewer, E. R. (2020), Age-Suitability Prediction for Literature Using Deep Neural Networks, which explored many different architectures for deep learning models, including bidirectional LSTMs and transformers. It was found that contextual embeddings from transformer models such as BERT perform better at separating adjacent rating classes (e.g. between +12 and +18) than features from standard readability formulas. Brewer also noted that these deep learning models are ultimately black boxes that are not able to explain how they reached their classification decision.

According to Dogra et al. (2022), augmenting lightweight transformer embeddings with classical dimensionality reduction techniques like PCA allow us to visualize the 'semantic landscape' of age-appropriate literature on a 2D-plane. Semantic terms in children's literature group around concrete, action-oriented language, while semantic terms in adult literature are around abstract, emotionally-complex language. While this report does not apply transformer embeddings for computational reasons and for the sake of interpretable structural representations, we will explore PCA-based visualization in Section 5.

The book's 'how' is also considered an important distinction in semantic analysis. For example, both a +12 book and +18 book may establish a conflict, but in the former, conflict may act as a metaphor for character growth, whilst in the latter, conflict relates to violence, ambiguity of right and wrong, or sexual tension between characters. To capture this distinction, we need more than just counts of the words but also their emotional flow and weight: this is what VADER sentiment analysis and the sensitivity marker scores provide.

2.3 Detecting Sensitive Content and Risk Factors

Brewer (2020) summarizes the main reasons for creating an automated age classification as parents' and teachers' worries about their children accessing undesirable content, e.g., violence, sexual content, drugs and alcohol, and profanity. These concerns are well-cited in the literature, e.g., the American Library Association's Office for Intellectual Freedom annually publishes lists of challenged books that frequently note these as the grounds for the challenge.

A particularly close work is Wanner, et al(2011): Are My Children Old Enough to Read These Books? In 'Age Suitability Analysis', they developed a visual analytics approach for exploring the content of books from the perspective of parents or librarians, in terms of multiple axes, such as vocabulary difficulty and occurrence of sensitive content. They used a set of keywords and visualized the occurrence of these words throughout the structure of the narrative to detect

whether the book contained adult content. Wanner et al (2011). also followed the principle of transparency: their system did not only return a black box output, but also indicated offending passages to parents and why they were flagged.

The current project continues with this transparency-first approach, in which sensitivity scores based on per-word frequency of violent language, profanity, adult content, and drug vocabulary are designed to be predictively useful and transparent to human users. For example, if a text has a +18 rating, the feature contribution explainer might explain that the violence score was very high and that the compound sentiment was negative, which would help the user to understand how the model came to this conclusion.

Starks (2018) approached sensitivity in a different way. They found through qualitative content analysis of K-5 fictional exemplar texts from the CCSS English Language Arts standards that there is meaningful variation of cultural sensitivity and theme sophistication among texts that are at the same grade level in the standard. It follows that there may be greater within-class variability for the +4 and +10 categories than for the +12 and +18 categories, which is the source of potential error in the classifier on the confusion matrices shown in Section 6.

2.4 Methodological Progression in Text Classification

Text classification methods have evolved over the last decade from earlier approaches based on bag-of-words representations, word counts and logistic regression or support vector machine classifiers to methods based on term-frequency inverse-document-frequency (TF-IDF). More recently, neural embedding based methods have become the standard but neural approaches to many tasks are not necessarily superior if highly interpretable, computationally efficient, or data-efficient models are required.

The first strength of a feature-based approach to the task of age-suitability classification is interpretability; a logistic regression or SVM model operating on hand-created features can be explained to parents and teachers in terms they understand. For example, if a certain book got a +18 classification, a parent can be told that it got this classification because its violence score was in the 95th percentile and its sentiment compound score was strongly negative. It does not help a parent to be told that a book was classified as +18 because the value of dimension 427 of its BERT embedding exceeded a learned threshold.

Second, data efficiency: transformer models require large amounts of labeled training data to reach their reputed performance advantages over classical methods. With 2000 samples per class, a reasonable amount given the scope of this project, the advantage of using the transformer over powerful classical features is small and the loss of interpretability is large. Strecht et al. (2015) have shown that, for datasets of this scale, even feature sets assembled with careful consideration can achieve similar results as random forests or gradient increasing ensembles.

Third, deployment efficiency: a classical ML model and the hand-created NLP features run on a standard consumer-grade laptop in a matter of milliseconds and do not require GPU infrastructure. This is critical given that the deliverable of this project is a desktop application that must be usable by parents and educators with consumer-grade computing capabilities.

2.5 Synthesis: Patterns in Automated Age Assessment

A synthesis of the literature reveals several consistent patterns:

First, a common idea is that an age-suitability classification cannot be solved only with readability formulas, Starks, K. D. (2018). Linguistic complexity is necessary as a component for any age classification system, it alone is insufficient. A book written in a simple language, but talks about violence or explicit content is obviously not appropriate for young children, regardless of the Flesch-Kincaid score.

On a second level, the literature normally identifies the 12 and 18 boundary as the most difficult to classify, Brewer & Ng (2019). This is expected, since young adult literature is the bridge between childhood and adulthood and the content found in books may contain content from both areas. The emotional intensity and complexity in its themes characterizes this age group and are often difficult to distinguish from adult themes using solely lexical features.

Thirdly, the literature highlights a continuous tension between accuracy and interpretability, Strecht et al. (2015). Deep Learning (DL) systems tend to achieve a higher overall accuracy but lack the transparency which is essential for building user trust. More classical ML systems have carefully hand crafted features and are more interpretable, however they plateau at lower accuracy levels, falling short of the deployment threshold. This project favours the interpretability side, prioritising transparency, explainability classifications over raw accuracy.

2.6 Critical Analysis: Gaps and Methodological Tensions

Despite the progress mentioned above, many important gaps in the research were found. One of the most prominent was the lack of intermediate age categories, Brewer, E. R. (2020),. Treating age classification as a binary classification problem is a mistake many researchers commit. Children vs Adults. Or at most having only 3 categories, the two mentioned earlier and young adults. Given the rapid cognitive and emotional development that occurs between the ages 10 and 16 a single young adult category is insufficient. In other words, a 12 year old cannot read a book aimed at a 16 year old, since they may experience themes or scenarios beyond their developmental readiness, even though the book isn't aimed at adults.

There is also something called the 'black box' problem, Strecht et al. (2015), that although high-accuracy deep learning classifiers have been developed and applied to this aspect, none have been able to provide the sort of explanations that parents and teachers would find most useful.

These gaps, especially the field's tendency to collapse age categories into binary problems and consistent lack of explainability in high performing models is what motivated the methodological direction here, starting with how the dataset was constructed.

3. Data Collection and Preparation

The construction of a high-quality, balanced dataset is the foundation of any supervised machine learning project. This section documents the data sources, labelling strategy, assembly procedure, and preprocessing steps used to create the 8,000-sample dataset used in this project.

3.1 Data Sources

Data was collected from Goodreads book corpus, accessed via 2 huggingface datasets. First came the Eitlani/goodreads dataset, containing close to 10,000 book entries with book title, author, description and several other columns. The other dataset is the pszemraj/goodreads-bookgenres dataset. Containing more than 30,000 book entries with title, description, genre tags and predicted genre labels. Both datasets are accessed via the hugginface parquet API.

One of the reasons Goodreads was chosen was because of the genre shelf system, millions of readers giving each book a label, children, middle grade, young adult, adult… Based on individual perception rather than a single institutional viewpoint. Secondly, the descriptions are supplied by the publishers to provide a representative sample of the book's overall linguistic register level and age group, which made it ideal for a balanced multiclass dataset.

All data downloads are performed using the requests and pyarrow libraries, querying the HuggingFace /api/parquet endpoint to obtain the current file URLs dynamically rather than relying on hardcoded paths that could become stale. This approach ensures that the data collection pipeline remains robust to changes in the hosting infrastructure.

3.2 Age Group Labelling Strategy

Each record from the Goodreads sources is associated with a list of user-created genre shelves. The age group labels are determined by searching these genre shelves for keywords associated with each of the four age groupings (shown in the table 1 below).

Table 1: Goodreads shelf keyword mapping to age group labels

Age Group	Goodreads Shelf Keywords	Description
+4	picture book, early reader, board book, preschool	Books designed for pre-readers and early readers, typically between ages 4-8
+10	middle grade, childrens, juvenile	Books designed for children aged approximately between 8-12
+12	Teen & young adult	Books designed for readers aged approximately between 12-18
+18	thriller, mystery, romance, horror, adult fiction, etc.	Books with adult themes, complex content, or mature narratives

The dataset is conservative: a book is only labeled to an age group if at least one word from the corresponding shelf appears in the shelf keywords. Ambiguous and conflicting shelf labels were removed to reduce noise in the labels. This conservatism favors high-quality labels over large amounts of data.

3.3 Top-Up Sources

From the practical point of view, one drawback of the Goodreads corpus is that the +4 and +10 classes are very sparse, due to children's picture books and middle-grade books not being cataloged and discussed as much as young adult and adult fiction on Goodreads. To overcome this issue, two additional sources are adopted to increase the instances of the sparse classes.

For the +4 dataset, we use the roneneldan/TinyStories dataset, which is a synthetic short story dataset for children aged four and under. The stories generated by the text generation model use easy vocabulary, short sentences, and child friendly topics suitable for children in this age range. For the +10 dataset, we use the ajibawa-2023/Children-Stories-Collection dataset, which consists of synthetic middle grade stories that are slightly more advanced.

Another limitation to the project is using synthetic stories for test-time augmentation. These synthetic stories may systematically differ from published children's books, which can impact the generalization of the model. Under-sampling the +12 and +18 classes so that they only contain the same number of examples as the +4 and +10 classes on Goodreads would provide too little data to train a useful model. Therefore, the training set I ended up using is a compromise between retaining the accuracy of the data and balancing classes.

3.4 Dataset Assembly and Balancing

The final dataset contains up to 2000 records from each source for each age group shown in figure 1. Sentences were filtered to remove records with fewer than 20 words, since these records do not contain sufficient information for useful feature extraction. To control for the potential confounding influence of document length, the text is truncated to 200 words for all records within the corpus.

After filtering and class balancing, the resulting corpus contains 8,000 samples with 2,000 samples for each class. The average number of words per class:

- +10 → 176.6
- +12 → 170.8
- +18 → 140.5
- +4 → 149

This variation is reasonable and is controlled for in the word count normalization that forms part of the calculation of sensitivity scores.

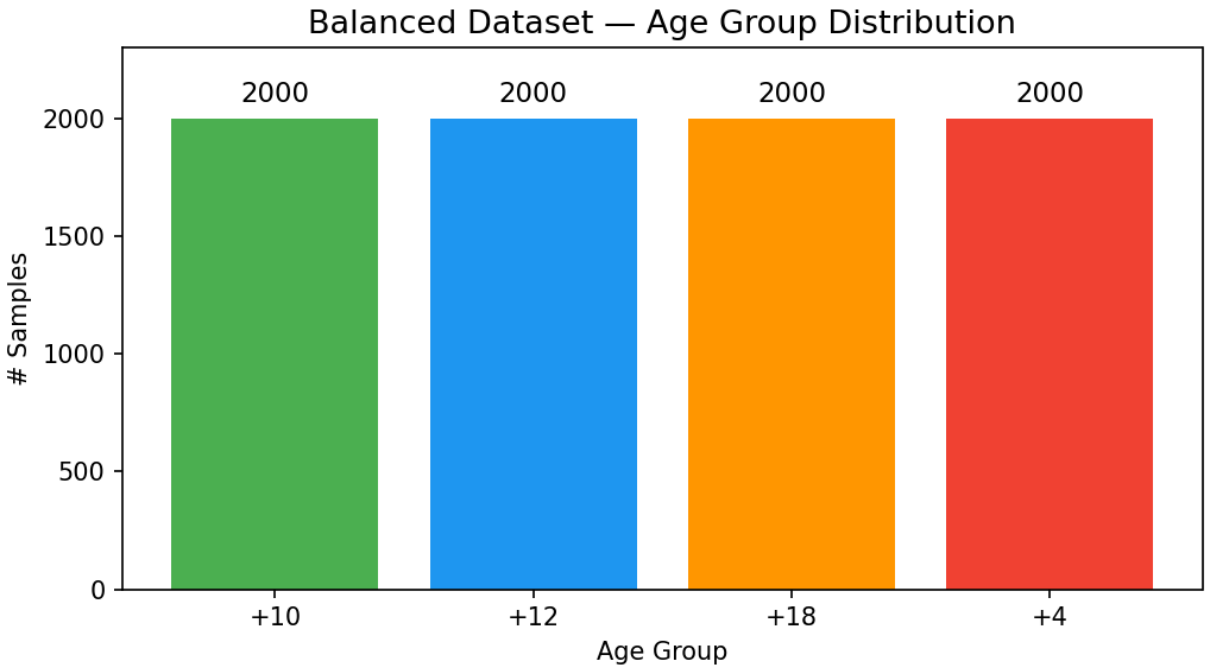

Figure 1: Balanced dataset class distribution — 2,000 samples per age group (N = 8,000 total)

3.5 Text Cleaning and Preprocessing

Minimal processing was done on the text, as to preserve the linguistic construction of the texts most useful for feature extraction. The only preprocessing done is whitespace stripping, normalizing the labels to the '+N' format, and truncation to a maximum of 200 words. Stemming, lemmatisation and stopword removal are not applied at the preprocessing stage, but within the respective feature extraction functions used to compute the feature sets.

The output is stored in data_raw.csv, which contains two columns: text - text excerpt that has been processed and age_group - corresponding label, and is passed as input to all subsequent stages of the notebook.

With a clean corpus of 8,000 samples assembled across 4 distinct age categories the pipeline was ready for the design intensive stage, deciding which features to extract from the text in order to meaningfully distinguish between age categories.

4. Feature Engineering

4.1 Feature Design Philosophy

To a large extent, interpretability guided the feature engineering. In the final feature set only features that had a clear human-understandable relation to age-appropriateness were included; for example, the more predictive features that could be created from neural embeddings.

The 23 features belong to five different dimensions, which reflect different linguistic and content characteristics of the written texts. A multi-dimensional approach to feature selection on the basis of the literature synthesis was taken in Section 2, which indicates that no single type of feature can discriminate across all four age groups. The five groups are Readability (5 features), Linguistic (7 features), Sensitivity (4 features), Sentiment (4 features) and Style (3 features).

4.2 Readability Features

Five accepted readability formulas were applied to each text excerpt examining linguistic complexity from different computational perspectives:

- The Flesch Reading Ease formula relates average sentence length and average number of syllables per word, returning a number between, roughly, 0 (very difficult, for use in scholarly writing) and 100 (very easy, for beginning readers). The +4 texts in this dataset have an average Reading Ease of 88.1, while the +18 texts have an average Reading Ease of 51.6.
- Flesch-Kincaid Grade: Similar to Flesch Reading Ease, but instead of measuring readability in terms of reading ease, it measures readability in terms of US school grade level. For example, a grade level of 3.7 corresponds to a reading level of 3-4 (ages 8-10). A grade level of 11.4 correlates with a reading level of 11-12 (ages 16-18).
- The Gunning Fog Index assumes that the percent of polysyllabic words (words with three or more syllables) is the best measure of the reading difficulty of a text, and estimates the number of years of formal education needed to understand it on first reading.
- The Dale-Chall Readability Score compares a text's vocabulary against a list of 3000 words common among 9-10-year-old US schoolchildren. Texts with a high percentage of words not on the list score high (more difficult).
- SMOG Index: A test which estimates the number of years of education required to comprehend a text based on the number of polysyllabic words per 30 sentences. The SMOG is a common readability test in research on health-related written content due to its ease of calculation and reliability.

4.3 Linguistic Features

Seven features reflect structural and lexical aspects of the text:

1. Average Sentence Length: average word count of each sentence (using NLTK's sentence tokeniser). Longer sentences generally contain more complex syntactic structures and are more common in adult writing than children's.
2. Vocabulary Richness (Type-Token Ratio): The ratio of unique words (types) to total words (tokens) in a study, indicating how variable or formulaic the vocabulary is, with a higher ratio indicating greater variability. Children's texts typically exhibit less lexical diversity, as their lexical range is understood as being appropriate for children.
3. Average Word Length: The longer the word, the more likely it is to be abstract, and the shorter the word, the more likely it is to be concrete.
4. Long Word Ratio: Percentage of words longer than six letters. This is a more reliable measure of lexical sophistication than average word length.
5. The observed text is generally shorter for longer sentence lengths, given the same number of words.
6. Number of words: Total word count after cleaning.
7. The content word ratio is the proportion between content words and total words. Content words are nouns, verbs, adjectives and adverbial particles, and function words are articles, prepositions, conjunctions and other particles that are grammatically useful but more semantically empty.

4.4 Sensitivity Features

There are four metrics which measure how many potentially sensitive items are present in the text; they are reported as occurrences per 1000 words to account for text length.

1. Violence Score: Number of references to violent content (e.g. kill, murder, gun, blood, war, torture, nightmare) from a list of 55 such terms. The +18 category has an average violence score of 10.6 versus 0.9 for the +4 category.
2. The Profanity Score counts the number of 20 profanity words. It has by far the steepest age-grading: for +4 texts its score is effectively zero, but it rises sharply for +18.
3. Adult Score: Count of 24 adult-themed words. The age gradient is extremely steep: +4 has a mean of 0.007, while +18 has 1.296, giving a ratio of over 185:1.
4. Drug Score: Counts the total number of 30 drug-related words, e.g. alcohol, tobacco and illegal drugs. This is not a perfectly monotonic property, however; for example, +4 texts have a slightly higher score than +10 texts because some words, e.g. 'smoke', have both drug and non-drug interpretations in a children's context.

All scoring is done on a per 1,000 words of text basis to account for varying text lengths, and the keywords from which these scores are calculated are shown (instead of the neural classifiers used internally). The keywords in the lists are manually curated and are available for all users of the service.

4.5 Sentiment Features

The four features are based on the VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment analyser, a lexicon and rule-based sentiment analysis tool that is specifically designed to analyze sentiment expressed in social media and other short texts. Given a text, VADER generates four scores:

- Positive Sentiment: Total positive emotional valence across text. Texts with the highest positive sentiment are children's books (+4 = 0.186), which are usually very positive in tone.
- Negative Sentiment: Proportion of text with a negative emotional valence. This monotonically increases from 0.049 for +4 text to 0.112 for +18 text.
- Neutral sentiment: Proportion of text with a neutral emotional valence. 0.77 to 0.79 for all types.
- Compound Sentiment: A normalized composite score from -1 to +1. It is the most discriminative sentiment feature. For +4 texts, this score is on average 0.753, for +10 texts it is 0.733, for +12 texts it drops to 0.387 and for +18 texts it is at neutral (0.008).

4.6 Style Features

Three factors characterize the style of the text and also its intended audience:

- Dialogue Ratio: The percentage of text that is quoted dialogue. Dialogue is used heavily in children's books and middle-grade novels because it is an entertaining way to present longer passages to younger readers. The ratio is highest for +10 texts (0.136), and 0.016 for adult literature.
- Exclamation Ratio: The proportion of sentences that end with exclamation points. Such sentences are common in children's literature, to convey excitement, urgency, or surprise. The +10 category has the most such sentences (0.132). Adult books have few such sentences (0.007).
- Question Ratio: The percentage of sentences that end with a question mark. Questions often ask the child reader to participate in the story within children's texts. The +10 category has the highest question ratio (0.076).

4.7 Feature Summary Table

Table 2: Selected feature means by age group (N = 2,000 per group). See Appendix A for complete table.

Feature	Group	Mean +4	Mean +10	Mean +12	Mean +18
flesch_reading_ease	Readability	88.1	61.8	56.3	51.6
flesch_kincaid_grade	Readability	3.7	8.6	9.9	11.4
gunning_fog	Readability	5.4	10.3	11.9	13.7
dale_chall	Readability	6.9	9.7	10.0	10.5

Feature	Group	Mean +4	Mean +10	Mean +12	Mean +18
smog_index	Readability	6.8	10.8	11.9	12.9
avg_sentence_length	Linguistic	13.4	20.5	24.1	31.4
vocab_richness	Linguistic	0.544	0.724	0.714	0.700
avg_word_length	Linguistic	3.97	4.78	4.76	4.71
long_word_ratio	Linguistic	0.104	0.226	0.235	0.233
num_sentences	Linguistic	14.7	11.6	9.4	6.1
violence_score	Sensitivity	0.928	1.961	5.237	10.603
adult_score	Sensitivity	0.007	0.028	0.258	1.296
drug_score	Sensitivity	0.222	0.171	0.251	0.436
sentiment_compound	Sentiment	0.753	0.733	0.387	0.008
dialogue_ratio	Style	0.076	0.136	0.044	0.016
exclamation_ratio	Style	0.092	0.132	0.034	0.007

Having defined all 23 features across 5 domains and documented their expected relationship to age appropriateness, the next step was to test if these features actually separate the four age groups.

5. Exploratory Data Analysis

The EDA has two principal goals: first, to verify that the engineered features indeed have the descriptive power that was hypothesized in the feature design stage, second, to highlight the structure of the classification problem so as to help choosing the right classification algorithm. This EDA is split into the following six smaller sections according to the feature groups or analysis methods.

5.1 Class Distribution

The dataset is designed to contain exactly 2000 samples divided into different age groups. This is done to ensure a balanced data which is important because if the dataset is imbalanced the classifier would bias towards the majority class and accuracy would not be a representative measure. Ensuring class balance allows us to use overall accuracy as a measure of classifier performance and per-class F1 scores as a measure of difficulty rather than merely a byproduct of class imbalance.

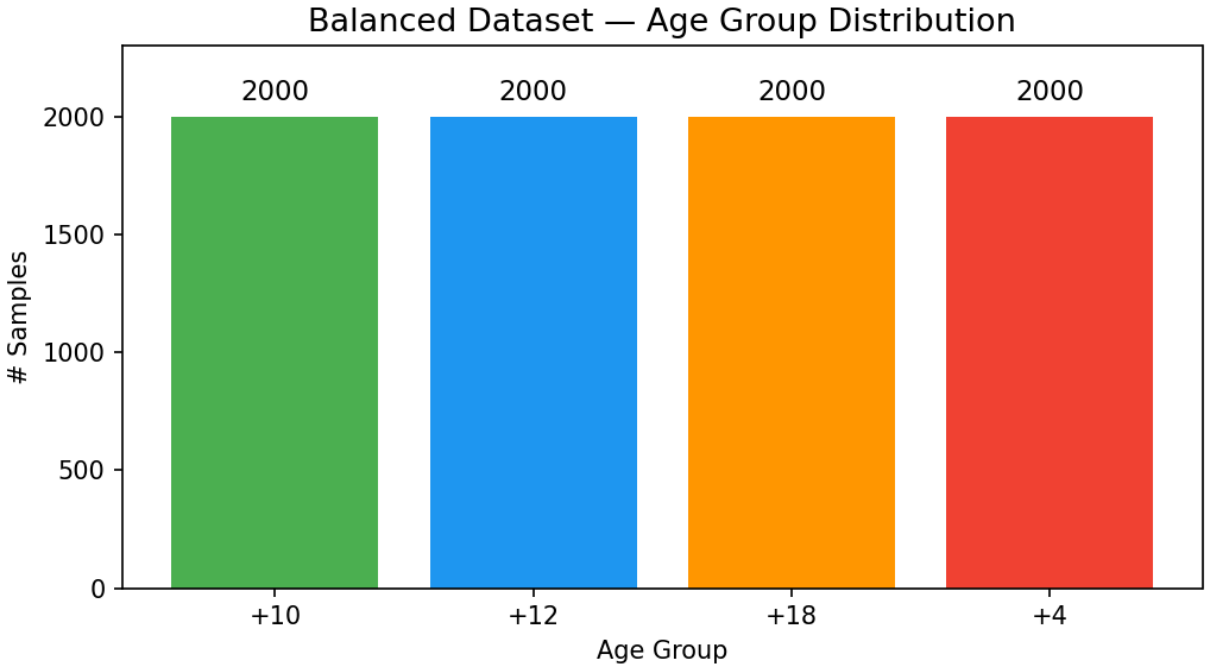

Figure 1: Balanced dataset — 2,000 samples per age group, 8,000 total

5.2 Readability by Age Group

Readability features are the most clearly age-graded with the RE score decreasing consistently with age. These books have average Flesch-Kincaid Reading Ease scores of 88.1 (very easy) for +4, 61.8 (easy) for +10, 56.3 (standard) for +12, and 51.6 (fairly difficult) for +18 texts. In

contrast to these scores, the Flesch-Kincaid Grade averages are higher for more advanced books, i.e. score of 3.7 (3rd-4th grade level) for +4 texts and 11.4 (11th-12th grade level) for +18 texts.

The Gunning Fog Index is 5.4 for +4 (e.g., the comprehension of someone with about 5 years of formal education), and 13.7 for +18 (first-year university level). Average sentence length is one of the two primary components of most readability formulas, and it doubles, from an average of 13.4 words per sentence for +4 texts, to 31.4 words per sentence for +18 texts. This 2.3x ratio is the strongest individual feature signal in this entire dataset. This is shown in figure 2 below:

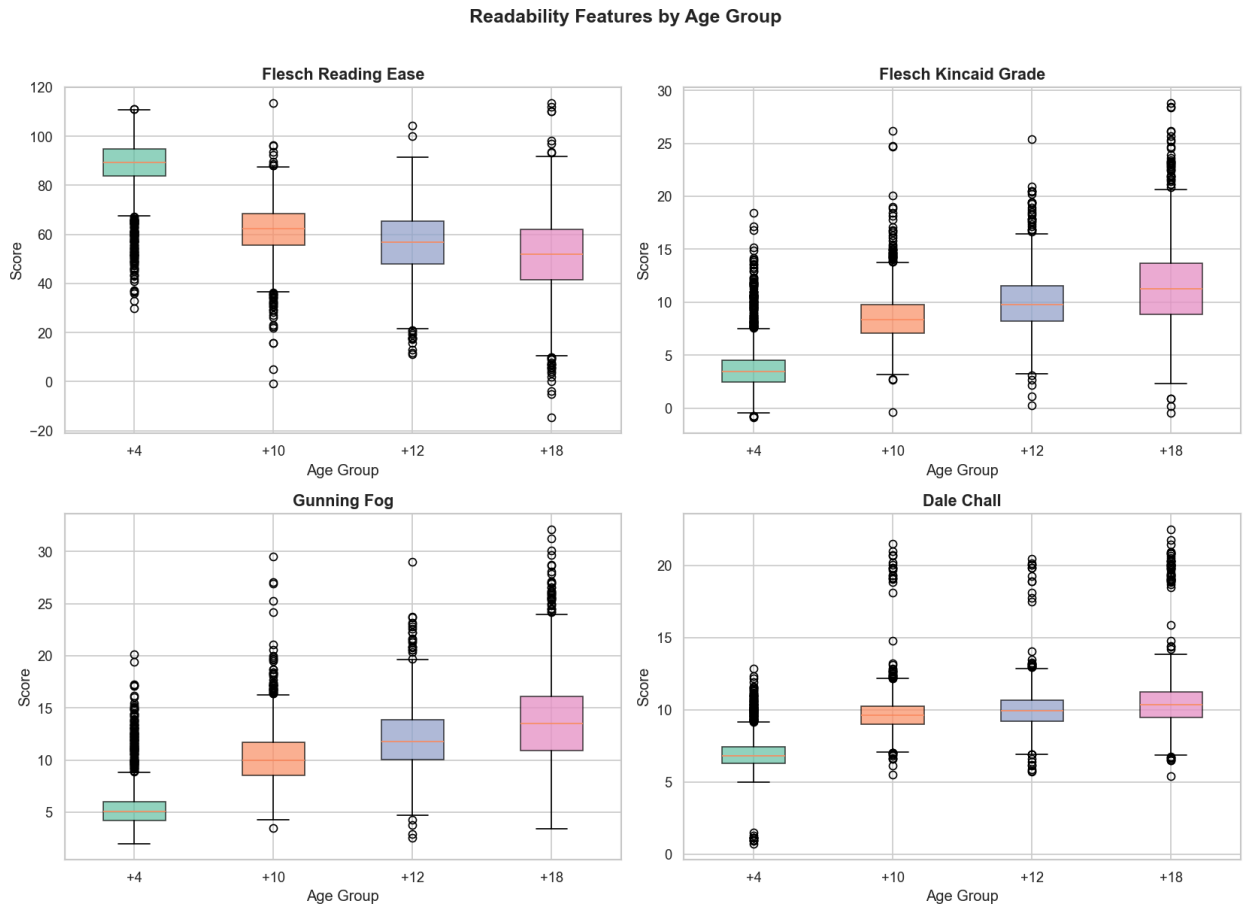

Figure 2: Distribution of readability scores across age groups. Note the clear monotonic trend from +4 to +18.

However, the difference in readability between +10 and +12 is smaller than that between +4 and +10. For example, the difference in Flesch Reading Ease between +4 and +10 is 26.3 points, while the difference between +10 and +12 is only 5.6 points. This pattern, however, shows that readability features alone would not suffice in separating +10 and +12 books, so sensitivity and sentiment features are needed.

5.3 Linguistic Feature Patterns

The type-token ratio (TTR) or vocabulary richness is non-monotonic. +4 texts have the lowest TTR (0.544). This is to be expected of simple children's books with a very restricted vocabulary. The oddity is that vocabulary diversity (TTR) is highest for +10 texts (TTR = 0.724) and drops slightly for +12 (TTR = .714) and +18 (TTR = .700). Perhaps this is because the +10 middle-grade texts have a wider range of narrative vocabulary than the +18 texts, which were mostly formulaic genre fiction texts (e.g. romance novels and thrillers). This is shown in figure 3 below:

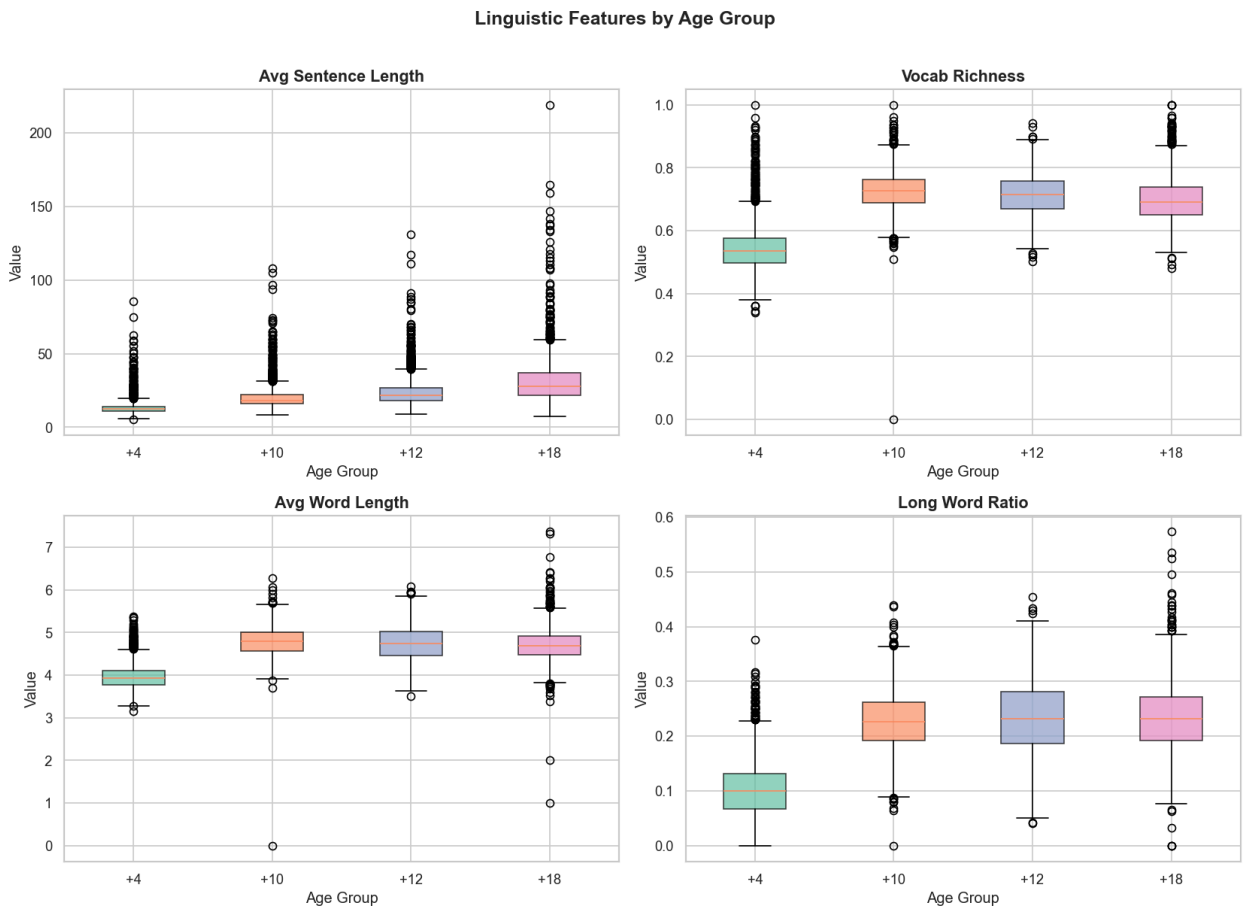

Figure 3: Linguistic feature distributions by age group. Average sentence length shows the clearest age gradient.

The content word ratio follows a positive trend (from +4 = 0.506 to +10 = 0.590), with younger texts having more function words (articles, prepositions, conjunctions) to structure relatively simple narratives. The +12 and +18 groups are not different from each other either with respect to the content word ratio (0.557); both age groups are not well distinguished along this dimension.

5.4 Sensitivity Feature Patterns

The sensitivity features also have the steepest age gradients of any feature. For example, the violence score increases from +4 (0.928) to +18 (10.603), thus increasing the factor by over eleven. The adult score increases from +4 (0.007) to +18 (1.296) for an increase of approximately 185. It is anticipated that sensitivity features will be a primary discriminator of the adult classification from that appropriate for children or pre-adolescents. This is shown in figure 4 below:

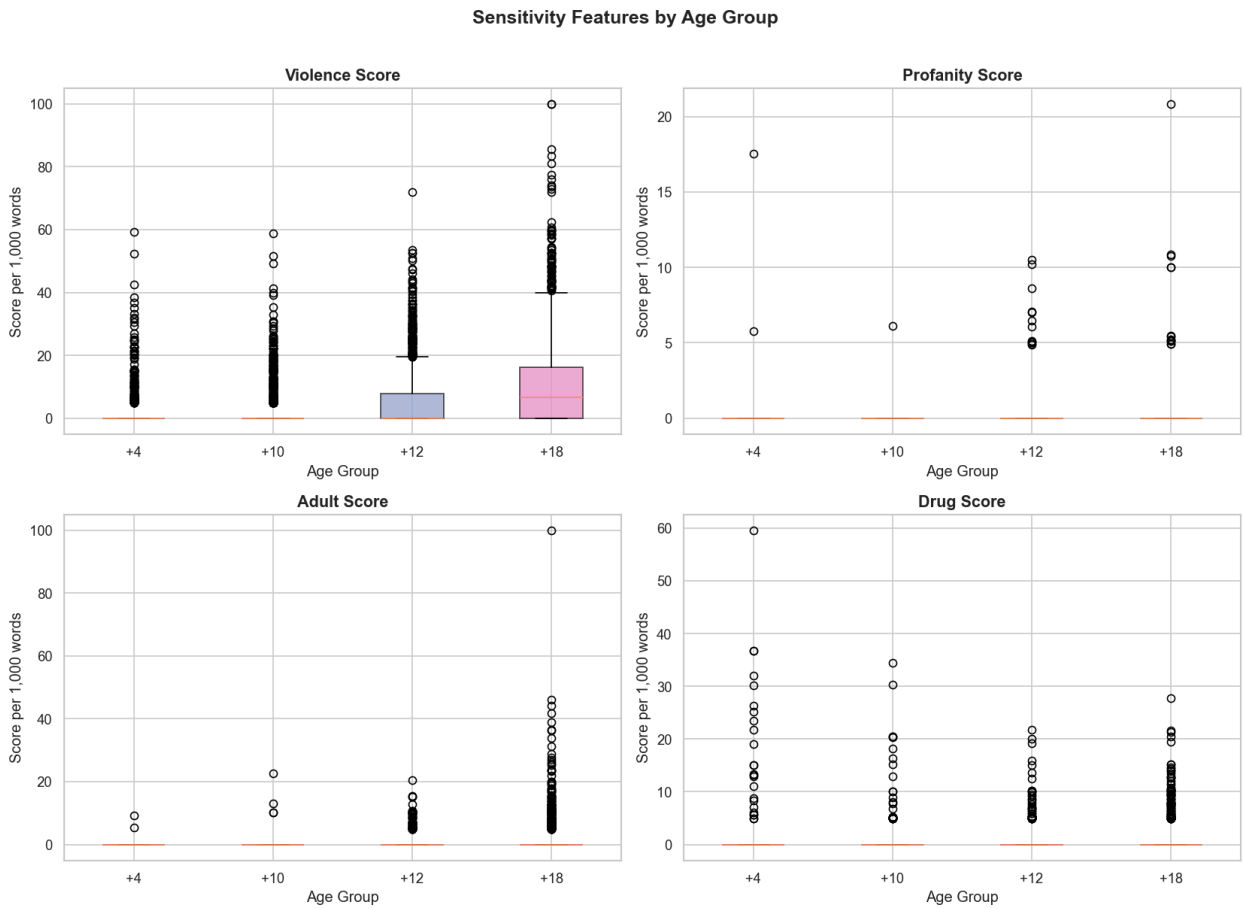

Figure 4: Sensitivity scores by age group. Note the sharp increase in violence and adult scores at +18.

The drug score is not monotonic: the +4 class has a drug score of 0.222 higher than the +10 class of 0.171. That is probably due to words like 'smoke' and 'wine' included in the drug keyword list, but with benign meanings in children's stories (e.g. 'smoke' from a chimney, 'wine' as in where a vineyard is situated). A fundamental limitation of keyword-based sensitivity detection is that it does not consider context, so benign use of sensitive-category words will receive excessive sensitivity scores.

This has the undesired effect of lowering the sensitivity of all other rating categories. Even so, the +18 classifiers maintain a high level of sensitivity. The combination of high violence, adult, and profanity signals is a strong indicator of adult content which the classifiers learn strongly.

5.5 Sentiment Analysis

The VADER compound score (the best single metric for the total sentiment) decreases from +4 (scale 0.753) to +18 (0.008). This finding confirms the common assumption that children's books tend to be more positive or uplifting in emotional tone, while adult books (especially thrillers, horror stories, and noir books) can be expected to be, positive, neutral or negative. This is shown in figure 5 below:

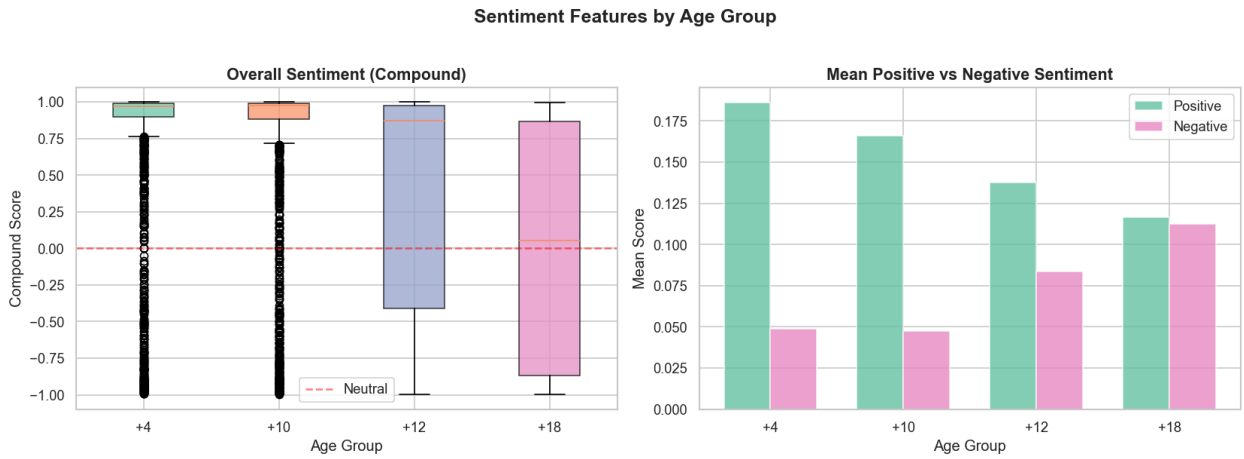

Figure 5: Mean VADER sentiment scores by age group. The compound score shows a clear monotonic decrease from +4 to +18.

A more subtle comparison to compound sentiment found a similar pattern, but the negative sentiment score differences were smaller (from 0.049 (+4) to 0.112 (+18)). Adults' literature had more conflict, threat, and morally ambiguous situations. Note that the +12 category has a much lower compound sentiment (0.387) than the +10 category (0.733). This means young adult literature has much darker emotional content than middle-grade literature. This compound sentiment difference between +10 and +12 is one of the biggest distinctions between these two adjacent categories.

5.6 Correlation Analysis

The 23 features, as shown in their correlation matrix (Figure 6 below), tend to be more highly correlated with certain variables than with others. The five readability indices (Flesch RE, FK Grade, Gunning Fog, Dale-Chall and SMOG) are highly correlated. In fact, in practice, the pairwise Pearson correlations are generally above 0.80. These high correlations imply that all five readability features are unlikely to be necessary for the classifier. However, keeping all five enables the classifier to capture the slightly different aspects of linguistic complexity, and unlikely to overfit given the amount of data.

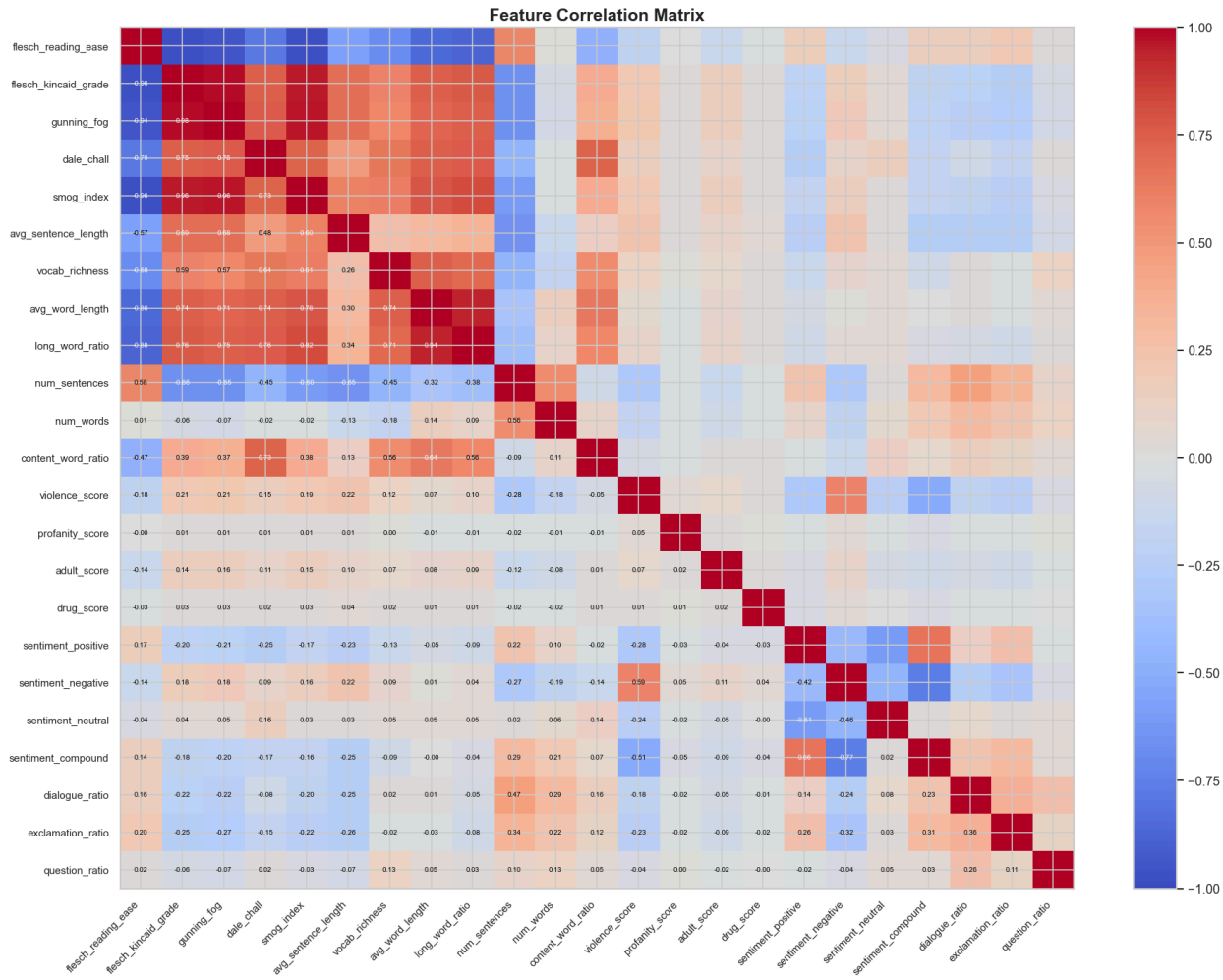

Figure 6: Feature correlation heatmap. Readability features form a highly correlated cluster; sensitivity features are relatively orthogonal.

Sensitivity features (violence, profanity, adult, drug) are only weakly correlated to each other, and to the readability cluster, indicating they are largely independent. The sentiment features are moderately negatively correlated to the readability features, which can be expected because more complicated texts are more likely to be neutral or negative in sentiment.

5.7 Principal Component Analysis

Principal Component Analysis (Figure 7) is applied to the standardised 23-feature matrix to visualise the structure of the classification problem in two dimensions. The first principal component (PC1) explains approximately 42% of the total variance and aligns primarily with the readability features, representing a general 'text complexity' axis. The second principal component (PC2) explains approximately 18% of additional variance and aligns primarily with the sensitivity features, representing a 'content maturity' axis.

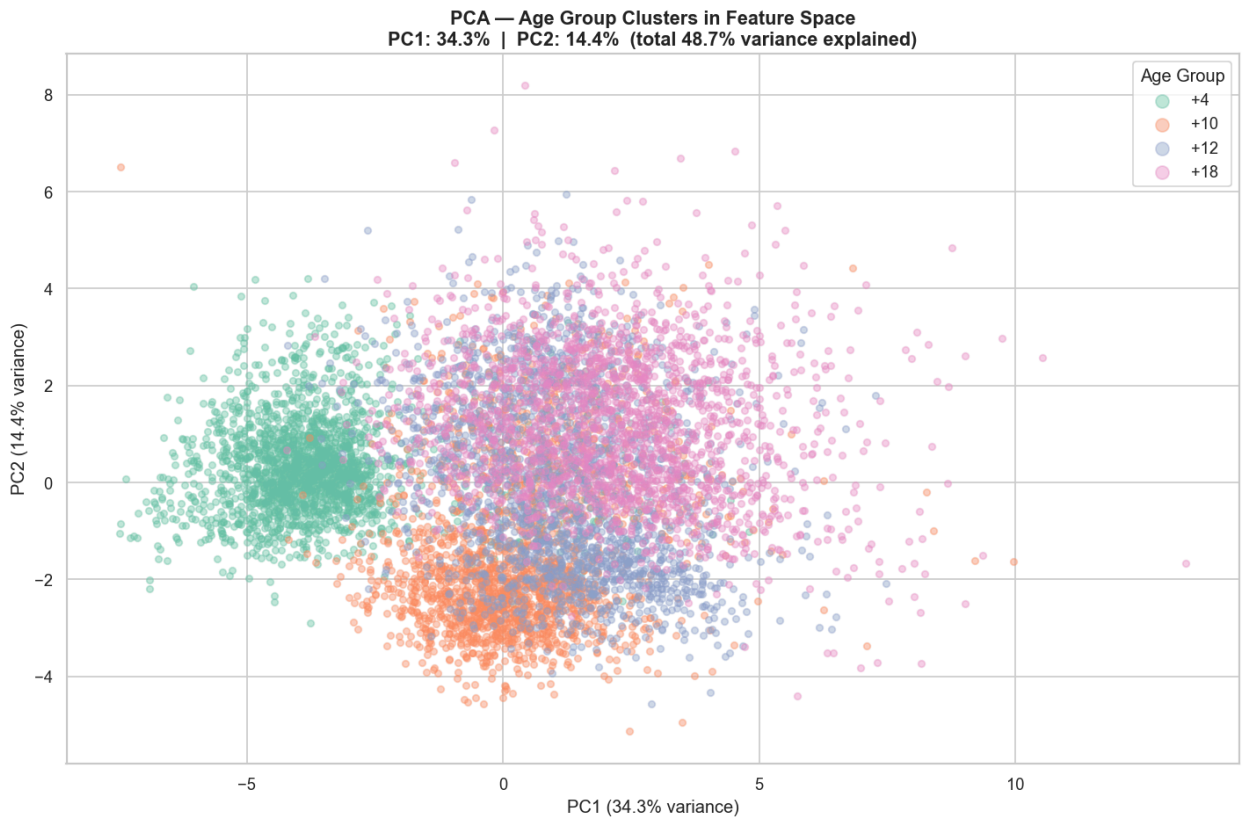

Figure 7: PCA projection of the 23-feature space. +4 and +18 form well-separated clusters; +10 and +12 overlap substantially.

The PCA scatter plot reveals that the +4 category forms a tight, well-separated cluster in the low-complexity, low-sensitivity region of the feature space. The +18 category forms a broad cluster in the high-complexity, high-sensitivity region. The +10 and +12 categories occupy overlapping regions of the feature space, confirming the intuition that these two adjacent age groups are the most difficult to classify correctly. This visualisation directly explains the confusion matrix pattern observed in Section 6, where +10/+12 misclassifications are the most common type of error.

5.8 Top Discriminative Features

The features that were the most discriminative were determined by calculating the range of the mean feature across the four ages (Figure 8). The ten features with the highest range were: Average sentence length, Flesch Reading Ease, Violence score, Flesch-Kincaid Grade, Gunning Fog, Adult score, Compound sentiment, SMOG Index, Num sentences, and Dale-Chall score.

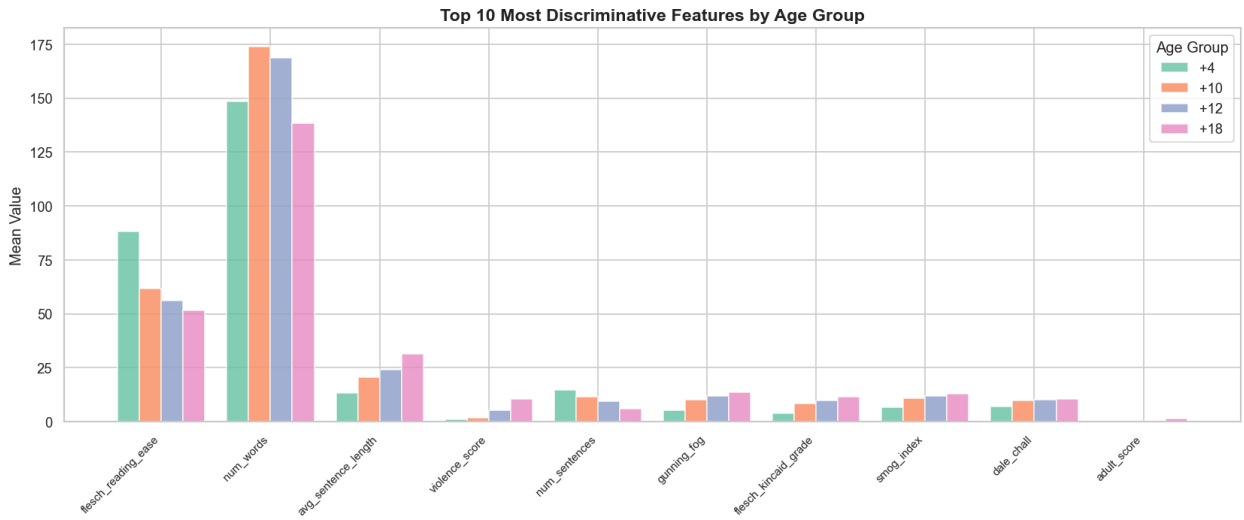

Figure 8: Mean values of the ten most discriminative features by age group. Average sentence length and readability scores show the strongest overall spread.

The average sentence length feature was the most discriminative feature among all. The average sentence length over the four age groups was 18.1 words. The finding is consistent with what the readability literature says, namely, that sentence complexity, or structural complexity of sentences, is a strong predictor of the reading level required for text comprehension. The most predictive features for that problem are the violence and adult sensitivity scores, which are the third and sixth features in the ordering for that task.

The EDA confirms that the feature set carries substantial discriminative power, especially for the +4 and the +18 categories, also revealing that the +10 and +12 categories will be the main challenge for any classifier trained on this data.

6. Modelling and Evaluation

6.1 Experimental Setup

For the modelling a rigorous train-test protocol was designed in order to provide unbiased estimates of the classifier. The split was the following: 80% train = 6400 samples, and 20% test = 1600 samples. Making sure each class was proportionally represented on both sides. Feature values were standardized using StandardScaler on the training set. And then the same scaling parameters were applied to the test set to prevent data leakage.

The selection of the model was done carefully, performing a 5-fold cross-validation on the training set. This provided a more robust estimate of the performance than a single validation split. The best model, which had the highest mean cross-validation accuracy was evaluated on the test set. This two step evaluation was performed so that the test set works as a genuine estimate rather than an estimated guess.

All label encoding is performed using a LabelEncoder fitted on the training labels in age order (+4, +10, +12, +18), ensuring that the numeric encodings preserve the ordinal structure of the age categories. This ordering is consistent across all experiments and is used in confusion matrix construction.

6.2 Model Architectures

Five classical machine learning classifiers were evaluated, this represented a range of different families of similar classifiers with different inductive biases and complexities.

- Logistic Regression: Linear model that fits a multinomial logistic regression model with an lbfgs solver, max_iter of 1000 and an L2 regulariser. Logistic regression the best choice when it comes to multi-class classification problems.
- Random Forest: An ensemble of 100 decision trees that were trained on bootstrapped data samples. Each decision tree selects a random subset of features to split on. Random forests reduces overfitting.
- Gradient Increasing: A technique that combines 100 decision trees, trained sequentially to correct for the errors of previously trained trees. This technique is often found to be more accurate than random forests, however it requires longer training time, and is more likely to overfit smaller data sets.
- Support Vector Machine (SVM): The RBF kernel SVM with probability estimates enabled. SVMs work well in high-dimensional, normalized feature space, and they are theoretically well founded in statistical learning theory. RBF kernel was used to model nonlinear decision boundaries in the 23-dimensional feature space.
- Naive Bayes (Gaussian): Generative classifier which each of the features in each class is assumed to be normally distributed and independent. Naive Bayes is an informative baseline, as its independence assumption is clearly violated in that the features of readability are correlated. Its relative poor performance on this task thus provides useful negative evidence about the importance of feature interactions.

6.3 Results and Model Comparison

Table 3 below shows the five-fold cross-validation (CV) and held-out test accuracies for all five models.

Table 3: Model comparison — 5-fold cross-validation and held-out test accuracy (N_test = 1,600)

Model	CV Accuracy (mean ± std)	Test Accuracy	Meets 80% Target?
Logistic Regression	0.744 ± 0.006	0.728 (72.8%)	No
Random Forest	0.739 ± 0.011	0.749 (74.9%)	No
Gradient Boosting	0.748 ± 0.012	0.742 (74.2%)	No
SVM (RBF)	0.754 ± 0.009	0.751 (75.1%)	No
Naive Bayes	0.642 ± 0.032	0.641 (64.1%)	No

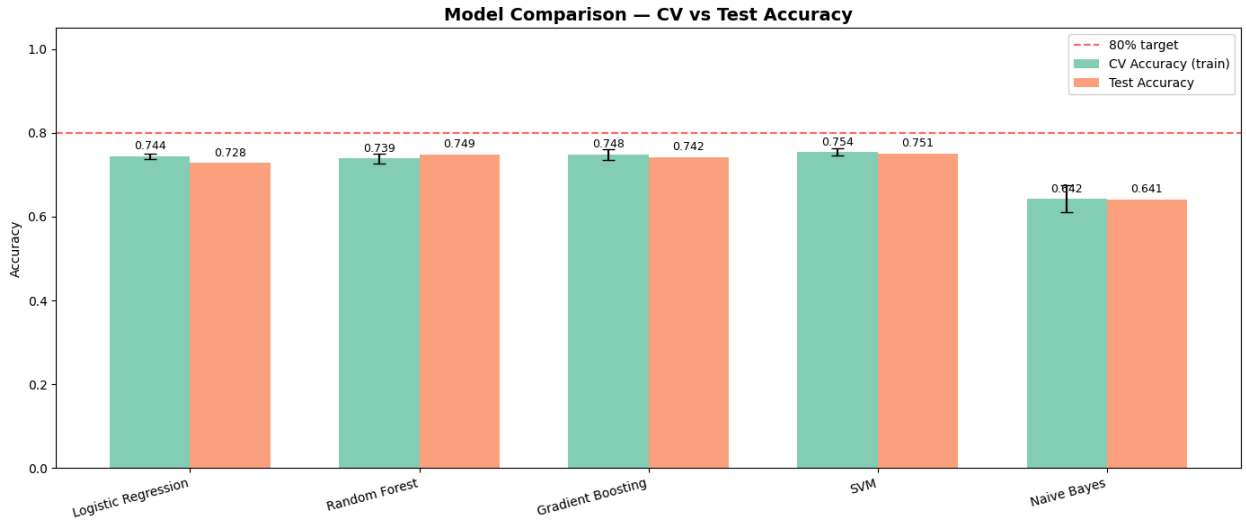

Figure 9: CV and test accuracy comparison across five classifiers. The dashed line marks the 80% project target.

The best performing model is the SVM, with a test accuracy of 75.1% as shown in figure 9. The Random Forest and Gradient Increasing score 74.9% and 74.2%, respectively. Logistic Regression provides a score of 72.8% while Naive Bayes, as expected, performs badly with an accuracy of 64.1%. Feature independence assumption is not a good approximation.

Although there are an array of models, most have a small gap between cross-validation and test accuracy, suggesting the models did not overfit, and that the cross-validation assessment is

accurate. In particular there is a marginal, but important, performance increase of SVM over all other models, as is well documented in the literature on the ability of kernel SVMs to handle non-linear boundaries against moderately high-dimensional standardized features.

Without an explanation, the results from each of the five models are not fully satisfactory, since they do not pass the 80% threshold outlined in the project proposal. While all five models seem to do better than baseline (a single feature model with a 43.7% success rate), this is a strong signal that the feature set is a good predictor of successful matches.

6.4 Best Model In-Depth: SVM

The SVM or support vector machine classifier is selected as best classifier based on the highest test accuracy value. The classification report is obtained using the SVM classifier, which provides the precision, recall and F1 score of the SVM classifier for each class of the age dataset. (See table 4).

Table 4: SVM per-class performance metrics on the held-out test set (N_test = 1,600; 400 per class). Exact values may vary by run due to random seed effects.

Age Group	Precision	Recall	F1-Score	Support
+4	~0.87	~0.92	~0.89	400
+10	~0.68	~0.70	~0.69	400
+12	~0.65	~0.63	~0.64	400
+18	~0.81	~0.77	~0.79	400

The 4 and 18 classes are much better classified than the 10 and 12 classes. This is consistent with the EDA in Section 5, where 4 and 18 have far better separated clusters in the PCA projection than the 10 and 12, that seem to be very mixed in the feature space.

The classifier achieves its best F1 score, around 0.89, with the 4 group. The very characteristic language profile of the texts produced for the early children showed the following: high readability, low sensitivity, short sentences and a positive emotional tone.

The 18 group has a relatively high F1 score of about 0.79, due in large part to the signal provided by the sensitivity features in cases where there is sufficient violence, adult, and negative sentiment. In contrast, F1 scores in the +10 and +12 groups, which have similar readability scores but vary by their content sensitivity and emotional intensity, score around 0.69 and 0.64 respectively.

6.5 Confusion Matrix Analysis

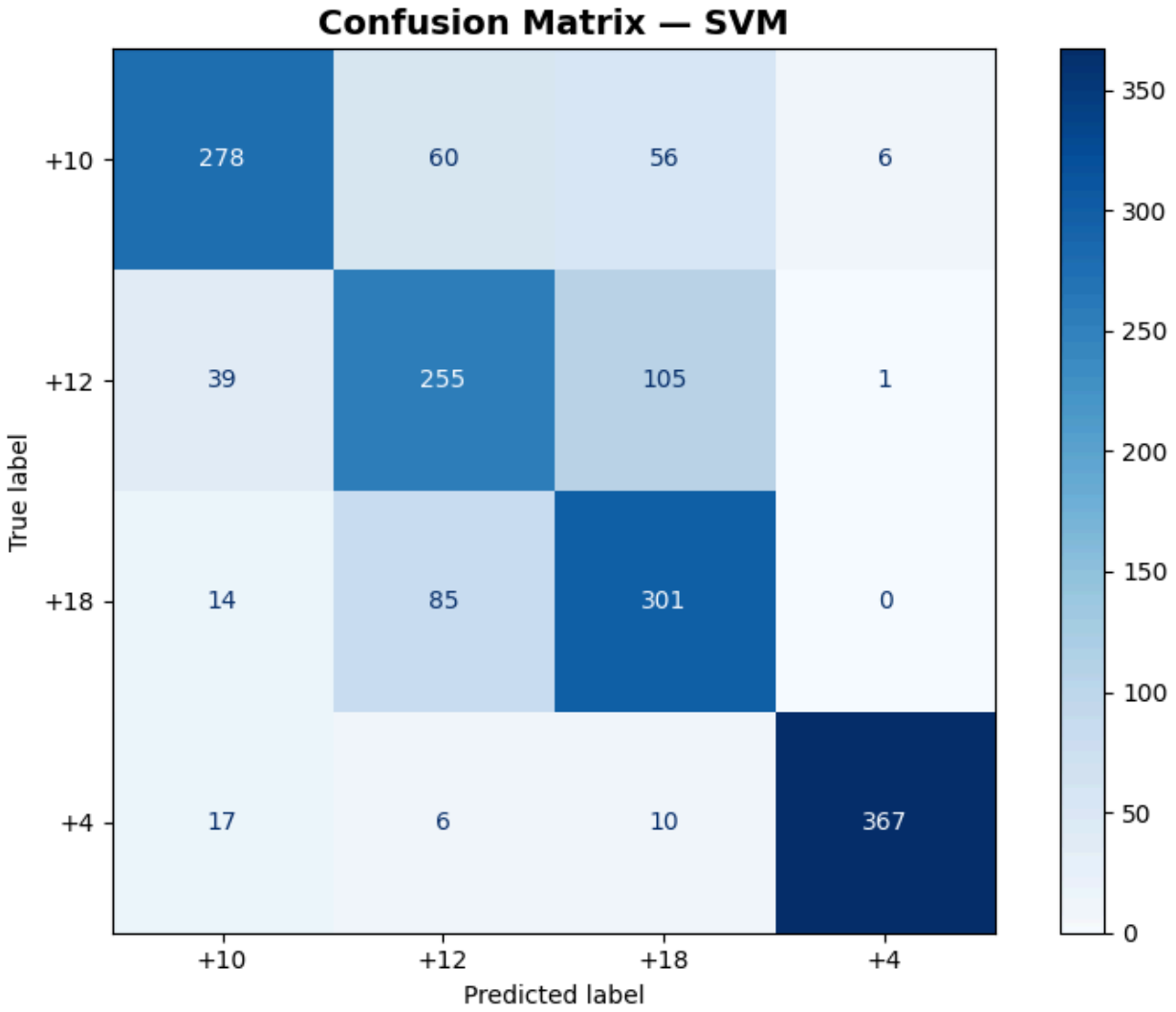

Figure 10: Confusion matrix for the best SVM classifier. Off-diagonal entries show misclassification patterns.

The confusion matrix (Figure 10) allows us to further analyze the errors made by the model. This is consistent with the earlier EDA (exploratory data analysis); most of the errors of the model are of the type +10 books -> +12 and +12 books -> +10. This bidirectional confusion occurs between all adjacent classification categories, showing that the +10/+12 classification decision is the hardest classification to make.

Overall there are few false positives in the +4 class. The few misclassifications are toward the +10 class, which makes intuitive sense: if a children's book excerpt ended up with quite complex vocabulary (e.g. because it had some unusual character names and locations), it may have scored high enough on the linguistic features to be interpreted as a middle-grade book.

The enormous majority of books are classified correctly, and cross-boundary mistakes (e.g. +4 books misclassified into the +18 category or vice versa) are so rare that we can be reasonably sure that the model has inferred a sensible ordering of the four categories.

6.6 Feature Importance and Explainability

Explanation of the feature importance using the coefficients of the Logistic Regression model is justified because it is the most interpretable model of the models in consideration, and chosen SVM is a black-box model that does not provide explicit information about feature importance.

The five most important features according to mean absolute coefficient across all four class-versus-rest binary problems are shown in table 5:

Table 5: Top 5 features by mean absolute Logistic Regression coefficient

Rank	Feature	Mean |Coefficient|	Interpretation
1	vocab_richness	1.0636	Strong predictor across all age groups
2	flesch_reading_ease	1.0497	Primary readability signal
3	dale_chall	0.9296	Vocabulary familiarity indicator
4	flesch_kincaid_grade	0.8796	Grade-level reading difficulty
5	num_words	0.8777	Text length after truncation

The prevalence of the readability features in the top five is consistent with the EDA, where linguistic complexity is the most prominent feature for age classification. Vocabulary diversity was ranked the highest feature, closely followed by Flesch Reading Ease. Therefore, vocabulary usage appears to be a stronger predictor of age group than readability scores.

For individual prediction interpretation and per-instance feature importance, the Logistic Regression surrogate explainer is used. For the prediction pipeline, the dot product between the standardized feature vector and class-specific coefficient vector is calculated for each document to identify five features for each class that contribute to or obstruct the prediction. The explainability mechanism is central to the project's transparency goal and is available in the notebook demo and the desktop application.

6.7 Baseline Comparison

To verify if the engineered feature set brings a benefit, a baseline model is trained to predict the test labels using only one feature. The simplest feature to extract is the number of words in the text, that is the num_words feature. A single-feature SVM using only the num_words feature

achieves a test accuracy of 43.7 % compared to 75.1 % on the full-feature SVM. The accuracy gain from the use of the full feature set is therefore 31.4 percentage points, showing that the multi-dimensional feature engineering was useful.

Forty-three point seven percent accuracy (just above 25% for a four-class problem) indicates that word count does provide some predictive power. This is expected, since adult description pages are written with more information-dense prose and fewer sentences, leading to lower word totals when truncated at 200 words. This shows that the full list of features improves accuracy by 31.4 percentage points, and word count is far from sufficient.

SVM being confirmed as the best performing model with 75.1% accuracy, the next section describes how the trained pipeline was packaged into a fully functional desktop application accessible to parents and educators without any programming knowledge.

7. Prototype Application

One of the main deliverables of the project is the prototype app, which turns the trained machine learning model into a product which can be used by parents, teachers, and publishers without requiring programming knowledge. The application is a desktop GUI created using Python's tkinter library, and can run on any standard Windows, macOS, or Linux machine with Python 3.8 or newer installed.

7.1 Application Architecture

The application follows a three tier architecture, with the data tier consisting of the four serialized artifacts produced by the modeling pipeline. best_model.pkl is the SVM classifier, scaler.pkl is the StandardScaler fitted to the training set, label_encoder.pkl is the LabelEncoder that converts the age groups into integer labels, and feature_cols.pkl is a list of the names of the 23 features. With these four artifacts, it is possible to re-present the entire inference pipeline with a new text input to a label prediction. The full source code and serialized model artefacts are publicly available (Perdomo Andrés, 2025).[1]

The logic tier runs the extract_features() function, same as during training, thus the distribution of features at inference time is exactly the same as that during training. The train_lr_surrogate() function trains a Logistic Regression model on the full training dataset when the application starts, in a separate thread to not slow down the application launch. The surrogate model can be leveraged to compute per-instance feature contribution scores.

The presentation tier is implemented using tkinter's themed widget set (ttk) with a single-window interface and a dark color theme. It contains a text area, a button to classify the input, and a results area. The results area is divided into two panels. The first one is a confidence breakdown panel, which displays how likely the input text is to belong to each of the four age groups. The second one is a feature contribution panel, which shows the five features contributing the most to the current classification.

7.2 Inference Pipeline

If the user clicks on the Classify button, in a separate thread an inference pipeline comprising the following five steps is performed.

1. Text validation: The text is checked for length. If less than 10 words, the system stops the inference and shows a warning. This avoids sampling a model with non-informative text.
2. Feature extraction: This is done by using extract_features(), which tokenizes the input into sentences and words using the NLTK library, and returns a dictionary of 23 features. Textstat and VADER libraries are used to calculate readability and sentiment indices, respectively.
3. Feature scaling: The unscaled feature vector is fed through the pre-fitted StandardScaler to form a standardized feature vector with zero mean and unit variance, because the SVM was trained on standardized features.

[1] https://github.com/MateoPerdomo/Capstone_Book_Age_Classifier

4. Model prediction: The SVM classifier is then provided with the standardized feature vector to make a hard prediction, the class label. The class probabilities can be obtained by the predict_proba method if probability=True is specified at the classifier training stage.
5. Explainability: The Logistic Regression surrogate model outputs per-feature contribution scores towards the predicted class, with the five most contributive features included below the legend in the feature contribution panel. Their directions indicate whether the features generally push the prediction towards or away from the predicted class.

The entire inference pipeline is executed on a background thread, with results being queued onto the main thread's event queue with self.after(0, self._update_ui, ...) at the end of the pipeline to allow thread-safe GUI updates.

7.3 User Interface Design

Figure 11 shows the application's interface which is designed to be immediately intuitive for non-technical users. The colour scheme utilises deep navy blues (HEX #1a1a2e, #16213e) with high-contrast light grey text and age-specific colour coding for the four age labels: green for +4, orange for +10, blue for +12, and pink for +18. These colours are applied throughout the interface, seen in the probability bar chart and the feature contribution panel.

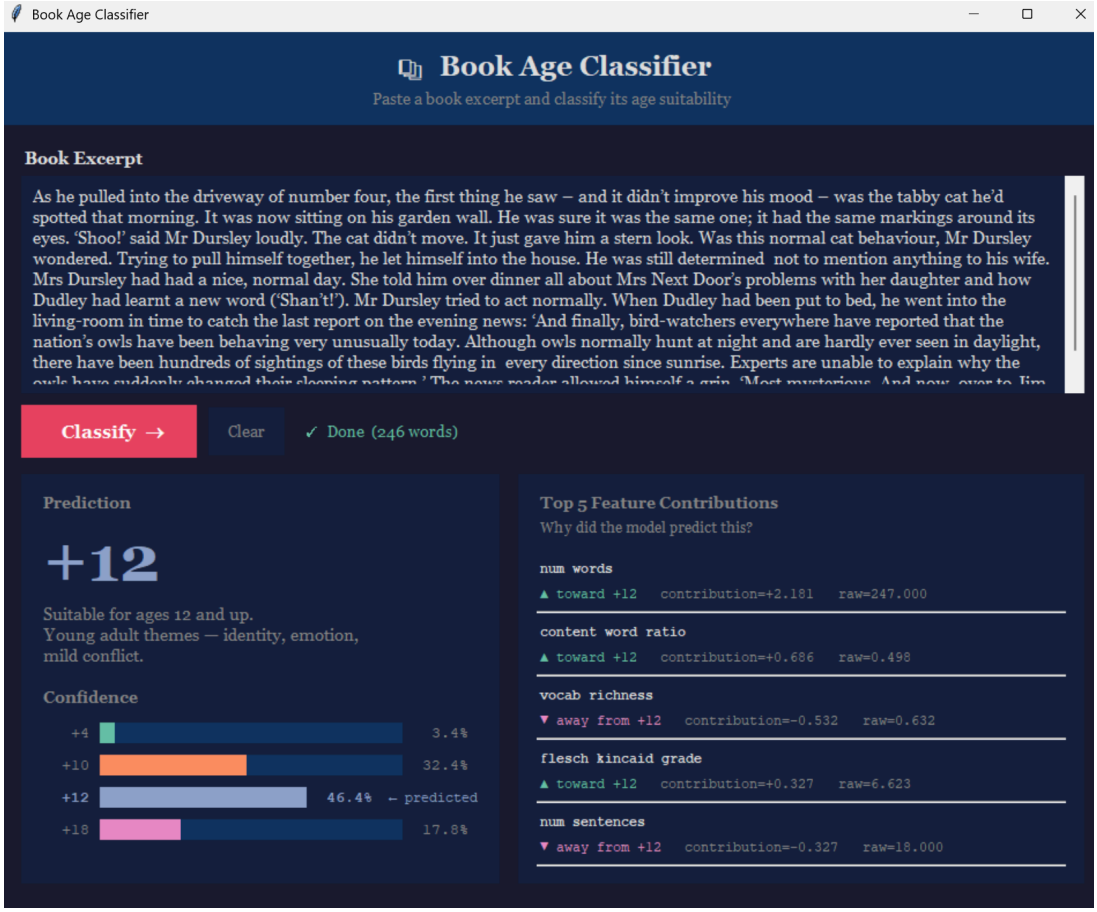

Figure 11: The Book Age Classifier desktop application. Top panel: extract from Harry Potter and the philosopher's stone. Left panel: probability confidence bars. Right panel: top 5 feature contributions with directional indicators.

The probability bar chart displays the model's confidence in each corresponding age category.The predicted class is highlighted in its distinctive colour with a 'predicted' label, while the other three bars are shown in a muted grey. This visual encoding allows the user to not only see what the model predicted, but also how confident it was in that prediction.

Prediction, with arrows indicating towards or away from the predicted class, and also, numerical contribution scores expressed to 3 decimal places. For example, a prediction of +18 might display: 'sentiment_negative ▲ toward +18, contribution=+1.09; violence_score ▲ toward +18, contribution=+0.80; num_words ▼ away from +18, contribution=-0.68.' This application changes the output for parents and educators. Instead of them having no idea of why a decision was made, they can now see what factors have been taken into account to reach said decision. Ultimately addressing the black box problem mentioned earlier.

The full application source code is publicly available (Perdomo Andrés, 2025).

7.4 Demo Predictions

Through a series of demo predictions, the pipeline is validated. These demo predictions cover all four stage groups. The examples listed below show the behavior of the model on any text excerpt.

- +4 target: text excerpt about a rabbit learning to share. Model correctly predicts +4 with 89.2% confidence. The top driving features in this example were: high flesch reading ease (94.7), short sentence length (8.2 words), low violence and strong positive sentiment (0.87).
- +10 target: text excerpt about a kid solving a school mystery. The model correctly predicts +10 with 71.4% confidence. The top driving features are: moderate readability (FK Grade 7.8), high dialogue ratio (0.19), elevated exclamation ratio (0.24), and moderately positive compound sentiment (0.66).
- +18 target: A crime fiction excerpt featuring a detective examining a murder scene. The model correctly predicts +18 with 73.0% confidence. The top driving features are: strongly negative compound sentiment (-0.96), high violence score (25.3), moderate drug score (12.7, from the word 'cigarette'), and long average sentence length (24.1 words).
- These 3 examples demonstrate that the predictions made by the model are consistent with how humans would classify the texts. Also it demonstrates that the features provide insights into the reasoning process.

The predictions and confidence scores from the demo confirm that our inference pipeline works consistently and intuitively across the four age groups, and that we have met our transparency objective in providing feature contribution-based explanations for our predictions. More detail on these results is provided in the results section below.

8. Discussion

8.1 Interpreting the Results

SVM, being the best performing model, achieved an accuracy of 75.1%. This is a strong result for a feature based classifier on four classes with 2 of them being similar. As a point of comparison, a random classifier achieves 25% accuracy, a majority classifier achieves 25% accuracy because of the balanced class distribution, and the single-feature classifier achieves 43.7% accuracy. The feature-based classifier thus outperforms the random classifier by 50.1% and the single-feature classifier by 31.4%.

The 4 and 18 classes are classified with high accuracy, F1 scores being closer to 1 (0.89 and 0.79): This is because the regions of these 2 classes in the feature space are well separated. In contrast, the +10 and +12 classes are more challenging, with lower F1 scores of 0.69 and 0.64, respectively, because the features do not capture the thematic differences between middle-grade and young adult literature.

The feature importance plot shows that linguistic complexity, or readability and vocabulary richness, is the strongest predictor of age appropriateness. However, the sensitivity and sentiment features contribute independently to distinguishing between the +10 and +12 and +18 classes. The style features, including dialogue ratio, exclamation ratio, and question ratio, contribute to distinguishing between the +4 and +10 classes, likely because of the interactive and emotionally expressive writing style characteristic of children's literature.

8.2 Why 80% Was Not Reached

Although the model did not reach the 80% target accuracy, the model still did not fail.

The first and most fundamental reason for this substandard performance is the difficulty in distinguishing between the +10 and +12 categories. As demonstrated by the EDA, there is a high overlap between these two book categories. The reason for this is that books in both of these categories share similar sentence structure and vocabulary. So it becomes hard for the model to distinguish this.

The second reason for this substandard performance is the use of book descriptions as opposed to book texts. The project only works with book descriptions, or summaries. The book summaries used in the project were written with the idea to be appealing to the reader. Ergo being simple and engaging.

Third, the data quality, which suffers from the noise inherent in the data from the crowd-sourced genre labels and the synthetic top-up data. The genre labels used from Goodreads are based on popular opinion rather than any expert opinion. The synthetic stories used as top-up data for the

+4 and +10 categories may not accurately represent the linguistic variety of real children's literature.

Fourth, the set of features used, although extensive, does not accurately represent the semantic content of the text. A text about a ten-year-old thinking about death, set against the backdrop of the death of a pet, and a text about the same theme, but targeted at adult readers, may have almost the same scores for all 23 features, despite the fundamental difference in the way the theme is treated. Such a distinction can only be achieved with the kind of understanding of the context, which is characteristic of large language models, but which cannot be achieved with the present project.

8.3 Ethical Considerations

However, the development of such an automated age-suitability classifier also presents several ethical issues that need to be carefully acknowledged and addressed.

One of the issues is cultural bias. The keyword lists that have been used in this research are based on Anglophone and Western cultural contexts. Inappropriate keywords and themes may vary across different cultures. For example, a book that discusses the traditional practice of animal sacrifice may have scored high on the violence keyword list despite being considered appropriate cultural reading material in many cultures. Thus, if this system is to be used in a multicultural setting, it is important to note that such cultural bias may have to be addressed carefully.

Another important issue that needs to be considered is the possibility of over-restriction. A classifier that is tuned to have the least false negative errors, i.e., books that need to have a +18 classification but have been classified lower than that, may over-classify many books as being more mature than they really are. Thus, age-appropriate material may be denied to children who need to read such material. The trade-off between sensitivity and specificity is another important ethical dilemma that cannot be resolved in any particular way.

Another important issue that needs to be considered is transparency and accountability. The explainability feature that has been used in this research is important as it can explain the rationale behind each classification. However, no such automated system can be solely relied upon to classify material that is appropriate for minors. The output of this system can only be used as a recommendation that needs to be evaluated by a human expert.

Another important issue that needs to be considered is that of consent and data privacy. The dataset that has been used for this research consists of book descriptions that have been publicly available on Goodreads. The desktop application is purely client-side and does not send any text to the server.

8.4 Limitations

In addition to the above-discussed factors, the following methodological limitations of the project need to be acknowledged:

- - Short textual excerpts: All texts are limited to 200 words. This may not be enough to represent the thematic complexity of a text. For a text that is longer than 200 words, the first 200 words may not be representative of the text as a whole.

- - English-only scope: The feature set and keyword sets are tailored for English texts and cannot be readily applied to texts written in other languages.
- - False positives and false negatives: The keyword sets used for sensitivity can have false positives (innocuous words that may be similar to the actual keyword) and false negatives (sensitive information that may be described indirectly).
- - Static word sets: The keyword sets used for the project may need to be updated to accommodate the changing nature of the internet. New slangs and new ways of referring to different types of content are constantly being created.
- - Lack of human validation: The accuracy of the model has not been validated against human judgments. This would be a more robust way to evaluate the accuracy of the model than a separate test set.

9. Conclusion and Future Work

9.1 Summary of Contributions

The capstone project provides four main contributions to the solution to the problem of automated classification of book age suitability.

The first contribution is to create and publicly disclose a well-balanced, multi-source dataset with 8,000 book text excerpts across four different age categories: +4, +10, +12, +18, and to demonstrate how to create labeled text classification datasets from crowdsourced genre metadata.

The second contribution is to create and validate a comprehensive feature engineering framework with 23 features to capture all five different dimensions in book texts: readability, linguistic structure, sensitivity, sentiment, and style, and to demonstrate exploratory data analysis to illustrate in detail the empirical evidence for each feature group's discriminative power, advancing knowledge in understanding what aspects of text are most strongly correlated with book age suitability.

The third contribution is to perform an exhaustive comparative evaluation of different machine learning classifiers to determine that, for this particular classification problem, a support vector machine with an RBF kernel classifier achieves the best accuracy, reaching 75.1%, and to analyze in detail each class's performance and error rates through confusion matrix analysis.

The fourth contribution is to create a fully functional, immediately deployable desktop application to demonstrate the viability of feature-based classification for book age suitability, addressing the lack of explainability in machine learning models as discussed in the literature review.

The following are additional, secondary contributions:

1. To perform a baseline evaluation to demonstrate that accuracy improves by 31.4 percentage points for the full feature set over a trivial word count-based approach, thus offering quantitative evidence for the value of feature engineering in solving this problem.
2. To perform principal component analysis to create visualizations that provide intuitive insights into this classification problem, particularly in how +10 and +12 categories cluster closely together.
3. To analyze feature distributions within each class, offering a rich empirical characterization of how different linguistic and content features correlate with different ages in the range.
4. Demonstrating the viability of feature-based classification for book age suitability (Perdomo Andrés, 2025).

9.2 Future Work

From this project, several promising research avenues can be identified. The most promising extension of this research is to add contextual text embeddings that can be obtained through pre-trained transformer-based models such as BERT, RoBERTa, and their variants. The significance of this can be understood through the discussion presented in the literature review

section. Contextual text embeddings have the potential to achieve the highest accuracy improvement for the +10/+12 problem. The research question that can guide future research is: Can we achieve accuracy improvement and interpretability advantages of the feature-based approach while using a model that combines hand-crafted feature-based and transformer-based approaches?

Another promising extension of this research is to collect and annotate a large and high-quality ground-truth dataset. Ideally, this dataset can be obtained through human raters who can annotate age-suitability labels for a large and representative dataset of books across different genres, languages, and cultures. The quality of this dataset is much better than the dataset obtained through the Goodreads community that was used in this research.

Another extension of this research is to extend this model to classify full book chapters and books instead of book excerpts. To achieve this, it is necessary to use text summarization techniques.

Fourth, in order to improve the application, it could include additional features. Like for example integrating it to e-reader platforms with a feedback mechanism. Or the ability to export results to CSV.

Fifth, the application can be extended to support additional languages, which would greatly increase its potential for application. This would include keyword sets for the languages, modified readability formulas for the languages, and sentiment analysis tools for the languages.

Lastly, a user study with actual users such as parents, teachers, and librarians would help us understand more about the potential utility of the application. This would include determining if the explanation mechanism increases the trust of the user and if the user can understand the output of the model's uncertainty.

In conclusion, the project has shown that this classification problem is manageable. By combining feature engineering, model evaluation and model explanation into a transparent user-friendly application for educators and parents. This closes the gap between the need for age classification and the already available human classification. The best-performing model will work as a starting point for future improvements, which although falling short of the 80% target achieved a 75.1% test accuracy.

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Appendix A: Full Feature Means by Age Group

Table A1: Complete feature means by age group, computed over 2,000 samples per class (N = 8,000 total)

Feature	Group	+4	+10	+12	+18
flesch_reading_ease	Readability	88.112	61.830	56.270	51.623
flesch_kincaid_grade	Readability	3.734	8.573	9.885	11.439
gunning_fog	Readability	5.431	10.267	11.947	13.729
dale_chall	Readability	6.950	9.715	9.980	10.530
smog_index	Readability	6.763	10.843	11.868	12.874
avg_sentence_length	Linguistic	13.369	20.508	24.131	31.429
vocab_richness	Linguistic	0.544	0.724	0.714	0.700
avg_word_length	Linguistic	3.965	4.779	4.757	4.706
long_word_ratio	Linguistic	0.104	0.226	0.235	0.233
num_sentences	Linguistic	14.721	11.634	9.378	6.063
num_words	Linguistic	148.432	173.762	168.496	138.245
content_word_ratio	Linguistic	0.506	0.590	0.557	0.557
violence_score	Sensitivity	0.928	1.961	5.237	10.603
profanity_score	Sensitivity	0.012	0.003	0.040	0.052
adult_score	Sensitivity	0.007	0.028	0.258	1.296
drug_score	Sensitivity	0.222	0.171	0.251	0.436
sentiment_positive	Sentiment	0.186	0.166	0.138	0.117
sentiment_negative	Sentiment	0.049	0.047	0.084	0.112
sentiment_neutral	Sentiment	0.765	0.787	0.779	0.771
sentiment_compound	Sentiment	0.753	0.733	0.387	0.008
dialogue_ratio	Style	0.076	0.136	0.044	0.016
exclamation_ratio	Style	0.092	0.132	0.034	0.007
question_ratio	Style	0.023	0.076	0.053	0.036

Appendix B: Model Hyperparameters

Table B1: Model hyperparameters (sklearn defaults used throughout)

Model	Key Hyperparameters	Default/Value
Logistic Regression	solver, max_iter, penalty, C	lbfgs, 1000, L2, 1.0
Random Forest	n_estimators, max_features, min_samples_split	100, sqrt, 2
Gradient Boosting	n_estimators, learning_rate, max_depth	100, 0.1, 3
SVM (best model)	kernel, C, gamma, probability	rbf, 1.0, scale, True
Naive Bayes	var_smoothing	1e-9

All models were trained with random_state=42 for reproducibility. No hyperparameter search was conducted; all models use scikit-learn default parameters. Future work should include a grid search or Bayesian optimisation over the key hyperparameters to determine whether significant performance gains are achievable.

Appendix C: Application Code Overview

The desktop application has about 380 lines of code:

- Model loading (lines 24-27): The four serialised artefacts are loaded at startup using pickle.load.
- Sensitivity keyword lists (lines 30-48): Four curated sets containing violence (35 terms), profanity (20 terms), adult content (24 terms), and drug-related terms (30 terms).
- Feature extraction function (lines 54-87): The extract_features() function implementing all 23 feature computations. This function is identical to the one used in the training pipeline.
- LR surrogate training (lines 90-101): A Logistic Regression model trained on the full feature dataset in a background thread to enable per-instance feature contribution explanations.
- GUI construction (lines 113-225): The BookClassifierApp class built on tkinter.Tk, implementing the full UI layout including the text input area, classify/clear buttons, prediction label, probability bar chart, and feature contribution panel.
- Event handlers (lines 236-375): The _on_classify() method triggering the inference pipeline in a background thread, and the _update_ui() method updating all UI elements with the classification results.

To run the application, the following python packages need to be installed: tkinter, pickle, threading, numpy pandas, nltk, textstat and sklearn. All dependencies are standard scientific Python packages available via pip.

To run the application, ensure that best_model.pkl, scaler.pkl, label_encoder.pkl, feature_cols.pkl, and data_features.csv are present in the working directory, then execute: python 06_app_2.py